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Selective alkane hydroxylation and alkene epoxidation using H₂O₂ and Fe(II) catalysts electrostatically attached to a fluorinated surface†

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Fe(II) complexes with pentadentate ligands, including *N*-heterocyclic carbene moieties, were prepared and electrostatically attached onto the perfluorinated surface of a mesoporous aluminosilicate. The heterogeneous catalysts were applied to the catalytic oxidation of cyclohexane and cyclohexene using H₂O₂ as an oxidant in CH₃CN, demonstrating high performance and selectivity in alkane hydroxylation and cyclohexene epoxidation.

The hydroxylation of alkanes as well as the epoxidation of alkenes using metal complexes as catalysts and hydrogen peroxide (H₂O₂) as a “green” oxidant should be of great importance in terms of the atom economy of oxidants without the production of hazardous waste.¹ However, in the use of H₂O₂ as an oxidant for metal-catalysed oxidation reactions, radical chain reactions are often involved through so-called Fenton reactions, making the reactions less selective.² In contrast, selective catalytic oxidation of organic compounds has been developed on the basis of the formation of high-valent Fe-oxo complexes as responsible species from the reaction of H₂O₂ with Fe(II) complexes;³ however, suppression of the over-oxidation of 2e[−]-oxidized products still needs to be considered.

In order to suppress the over-oxidation of 2e[−]-oxidized products, a hydrophobic micro-environment around metal complexes used as catalysts would be effective in achieving the efficiency and selectivity of the oxidation of hydrophobic compounds in polar media.⁴ In addition, in order to improve the durability of oxidation catalysts, it would be useful to fix the catalysts on the surface of a

support, making the catalyst heterogeneous to avoid decomposition due to intermolecular reaction between the reactive intermediate and the catalyst. Aluminosilicates, in which a portion of the Si atoms in the SiO₂ framework is replaced by Al, possess protons for charge compensation derived from the difference in the number of valence electrons between silicon and aluminum. Taking advantage of the fact that these protons can be exchangeable with other cationic species, aluminosilicates have been used as a robust support for converting cationic complexes that function as homogeneous catalysts into heterogeneous solid catalysts.⁵ So far, Fukuzumi and coworkers have reported the immobilization of metal complexes as catalysts in a mesoporous aluminosilicate, Al-MCM-41, to achieve the selective oxidation of benzene to phenol using H₂O₂ as an oxidant;⁶ however, no example has been reported on the introduction of a hydrophobic environment on the surface of aluminosilicates as a support for catalysts for the preparation of heterogeneous metal-complex catalysts. Thus, we have developed a heterogeneous catalyst consisting of Fe(II) complexes with pentadentate ligands with *N*-heterocyclic carbene (NHC) moieties as catalysts⁷ and a hydrophobized mesoporous aluminosilicate with fluorinated alkyl chains and negatively charged sites that interact electrostatically with positively charged catalysts to fix those catalysts.

Herein, we describe the selective hydroxylation of cyclohexane (CyH) and epoxidation of cyclohexene (Cyene) in water using aqueous H₂O₂ (30%) as an oxidant and Fe^{II}-NHC complexes immobilized on a hydrophobic fluorinated solid surface as catalysts. In this study, we have developed heterogeneous oxidation catalysts using PF₆[−] salts of Fe^{II}-NHC complexes, Fe^{II}-Cl[−]NHC,⁶ Fe^{II}-H[−]NHC and Fe^{II}-Me[−]NHC and an SBA-15 type mesoporous aluminosilicate bearing a hydrophobic surface decorated by perfluorinated C₆ chains, fluoroalkyl-functionalized Al-SBA-15. The three complexes are fixed on the surface based on electrostatic interaction to afford Fe^{II}-Cl[−]NHC@Al^F, Fe^{II}-H[−]NHC@Al^F and Fe^{II}-Me[−]NHC@Al^F, respectively, as catalysts for hydrocarbon oxidation (Fig. 1; abbreviations: Cl[−]NHC, 5,6-dichloro-1,3-dihydro-methyl-2*H*-benzimidazol-2-ylidene; H[−]NHC, 1,3-dihydro-1,3-bis

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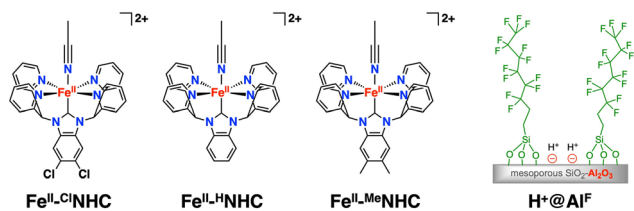


Fig. 1 Schematic representation of $\text{Fe}^{\text{II}}\text{-ClNHC}$, $\text{Fe}^{\text{II}}\text{-HNHC}$, $\text{Fe}^{\text{II}}\text{-MeNHC}$ and $\text{H}^+\text{@AlF}$. FC: $(\text{CF}_2)_5\text{CF}_3$.

(2-pyridyl)-methyl-2H-benzimidazol-2-ylidene; MeNHC , 5,6-dimethyl-1,3-dihydro-1,3-bis(2-pyridyl)-methyl-2H-benzimidazol-2-ylidene; $\text{H}^+\text{@AlF}$, $\text{C}_6\text{F}_{13}\text{-C}_2\text{H}_4$ -functionalized Al-SBA-15).

The syntheses of all complexes are described in the ESI†. The characterization of $\text{Fe}^{\text{II}}\text{-ClNHC}$ was described in our previous report.⁸ $\text{Fe}^{\text{II}}\text{-HNHC}$ and $\text{Fe}^{\text{II}}\text{-MeNHC}$ were characterized by ^1H NMR (Fig. S1 and S2 in the ESI†) and UV-vis absorption spectroscopy, ESI-TOF-MS spectrometry (Fig. S3 and S4 in the ESI†) and elemental analysis. The precursor of the hydrophobic support, Al-SBA-15, was prepared according to the procedure reported by Krishana and Selvam with some modifications.⁸ Its surface area, pore size distribution, and pore volume derived from the N_2 adsorption isotherm curves were consistent with the characteristics of an SBA-15-type mesoporous support (Fig. S5 in the ESI†). Anchoring the triethoxy-1H,1H,2H,2H-tridecafluoro-*n*-octylsilane on the surface of Al-SBA-15 yielded $\text{H}^+\text{@AlF}$. The hydrophobic nature of $\text{H}^+\text{@AlF}$ was evidenced by a large contact angle for H_2O (Fig. S6 in the ESI†). Immobilization of the Fe complexes was performed by cation-exchange of $\text{H}^+\text{@AlF}$ with PF_6^- salts of the $\text{Fe}^{\text{II}}\text{-NHC}$ complexes (Fig. 1) in acetonitrile. The details of synthetic processes are provided in the ESI†. The characterization of $\text{Fe}^{\text{II}}\text{-RNHC@AlF}$ ($\text{R} = \text{Cl}, \text{H}, \text{Me}$) was performed by UV-vis-NIR reflectance spectroscopy (Figs. S7–S9 in the ESI†). The loadings of Fe complexes on AlF for $\text{Fe}^{\text{II}}\text{-ClNHC@AlF}$, $\text{Fe}^{\text{II}}\text{-HNHC@AlF}$ and $\text{Fe}^{\text{II}}\text{-MeNHC@AlF}$ were calculated to be 7.3, 16, 23 nmol mg^{-1} , respectively, based on ^1H NMR spectroscopy. The details of the determination of the loading of Fe complexes are provided in the ESI†.⁹ The tendency for the amount of Fe complexes indicates that $\text{Fe}^{\text{II}}\text{-MeNHC@AlF}$ shows the largest loading amount among the three Fe complexes owing to the δ^- properties of the surface of AlF and the δ^+ properties of the surface of methyl substituents on the benzimidazole moiety.

Catalytic oxidation was performed in a glass tube containing CD_3CN (1.0 mL), $\text{Fe}^{\text{II}}\text{-RNHC@AlF}$ (10 nmol, $\text{R} = \text{Cl}, \text{H}, \text{Me}$), H_2O_2 as an oxidant (16 mM) and a substrate (100 mM) at 313 K under Ar atmosphere. The turnover numbers (TONs) after stirring for 3 h were determined by ^1H NMR spectroscopy. In the CyH oxidation, we could observe the formation of cyclohexanol (Cy-OH) and cyclohexanone (Cy=O) as two-electron-oxidised and four-electron-oxidised products, respectively. As a catalyst, $\text{Fe}^{\text{II}}\text{-ClNHC@AlF}$ showed the highest total TON of 744 among the three catalysts applied. We also performed CyH oxidation in trifluoroethanol instead of acetonitrile to investigate the effect of solvent; however, no oxidation product was detected (Table

Table 1 Catalytic oxidation of cyclohexane by H_2O_2 (30% aq) and $\text{Fe}^{\text{II}}\text{-RNHC@AlF}$ in CH_3CN under Ar

R	Product	TON ^a	Alcohol/Ketone
Cl	Cy-OH	705	18
	Cy=O	39	
H	Cy-OH	468	11
	Cy=O	43	
Me	Cy-OH	329	7
	Cy=O	50	

^a TONs = [Product]/[catalyst]. Conditions: catalyst: 10 nmol, $[\text{H}_2\text{O}_2] = 16$ mM, [substrate] = 100 mM, $T = 313$ K, and reaction time = 3 h. TONs were determined by GC-MS spectroscopy. No side-reaction products apart from those listed in the table were observed.

S1, ESI†). This result indicates that a protic solvent is not suitable for the oxidation reactions reported herein. The high alcohol/ketone (A/K) ratios indicate that $\text{Fe}^{\text{II}}\text{-RNHC@AlF}$ selectively catalyses the $2e^-$ -oxidation of CyH by suppression of over-oxidation owing to the hydrophobic fluorinated surface of AlF . Among the three catalysts, $\text{Fe}^{\text{II}}\text{-ClNHC@AlF}$ showed 95% selectivity. It should be noted that $\text{Fe}^{\text{II}}\text{-RNHC}$ complexes as homogeneous catalysts and $\text{H}^+\text{@AlF}$ as the support showed 100-fold lower catalytic activity and no reactivity in CyH oxidation with H_2O_2 (Table S2, ESI†).

To gain a mechanistic insight into the catalytic C-H oxidation using $\text{Fe}^{\text{II}}\text{-ClNHC@AlF}$, we employed cumene, toluene and cyclooctane as substrates. The total TONs (3 h) of the oxidation reaction for cumene, toluene and cyclooctane were 1430, 1155, 855 (Table 1 and Table S3, ESI†). The A/K ratios for toluene, cyclooctane were 20 and 23, which are comparable to that (18) of CyH (Table 1 and Table S3, ESI†). We also employed *cis*-1,2-dimethylcyclohexane as a substrate to investigate the stereochemistry of the product upon hydroxylation in an $\text{Fe}^{\text{IV}}\text{=O}$ -mediated mechanism. The oxidation of *cis*-1,2-dimethylcyclohexane afforded only *cis*-1,2-dimethylcyclohexane-1-ol with complete stereo-retention with a TON of 420.¹⁰ The stereo-retention is more complete than that reported for high-valent nonheme Fe=O complexes.^{3a} These results indicate that the catalytic oxidation reaction using $\text{Fe}^{\text{II}}\text{-RNHC@AlF}$ as a catalyst and H_2O_2 as an oxidant in acetonitrile should be conducted not by free radicals derived from H_2O_2 but by $\text{Fe}^{\text{IV}}\text{=O}$ intermediates. In addition, we performed a competitive catalytic oxidation reaction using an equimolar mixture of cyclooctane and Cy-OH (50 mM each) as a substrate. The oxidation of the mixture afforded mainly cyclooctanol with a total TON of 780 (Table S4, ESI†), which is comparable to that (855) of cyclooctane oxidation (Table S3, ESI†); Cy=O was also obtained as a minor product from Cy-OH with a low TON of 20. This clearly indicates that alkane oxidation is preferred to alcohol oxidation in our catalytic system owing to the hydrophobic surface of AlF even if Cy-OH has a lower $\text{BDE}_{\text{C-H}}$ than cyclooctane. High alcohol selectivity has been reported for C-H oxidation in strong hydrogen-bond-donor solvents, such as fluoroalcohols which deactivate alcohols.¹¹ In contrast, our catalytic system has achieved high alcohol selectivity even in CH_3CN owing to the hydrophobic micro-environment where

hydrophobic alkanes are favourably incorporated and hydrophilic alcohols are eliminated.

To investigate the durability of $\text{Fe}^{\text{II}}\text{-}^{\text{R}}\text{NHC@Al}^{\text{F}}$ catalysts, a recycling test for the heterogeneous $\text{Fe}(\text{II})$ catalysts in CyH oxidation was examined under the same conditions. After the 1st catalytic CyH oxidation reaction, the solid catalyst was collected by centrifugation and washed with acetonitrile five times. As for the 2nd–5th catalytic CyH oxidation reactions, each run was performed to determine the amount of each product by GC analysis after the determination of the amount of the corresponding Fe catalyst retained on the surface by UV-vis-NIR reflectance spectroscopy. The results of recycling tests for the three $\text{Fe}^{\text{II}}\text{-}^{\text{R}}\text{NHC@Al}^{\text{F}}$ catalysts in CyH oxidation reactions are depicted in Fig. 2. The catalytic activity of $\text{Fe}^{\text{II}}\text{-}^{\text{Cl}}\text{NHC@Al}^{\text{F}}$ and $\text{Fe}^{\text{II}}\text{-}^{\text{H}}\text{NHC@Al}^{\text{F}}$ was gradually decreased with each cycle of catalytic oxidation (Fig. 2a and b). In addition, the UV-vis-NIR reflectance spectra of the two catalysts showed the disappearance of absorption derived from the corresponding $\text{Fe}(\text{II})$ complexes (Fig. S10 and S11 in the ESI†). It is noteworthy that the catalytic activity of $\text{Fe}^{\text{II}}\text{-}^{\text{Me}}\text{NHC@Al}^{\text{F}}$ remains up to the 5th run (Fig. 2c). The total TONs relate to the amount of Fe complexes based on the UV-reflectance spectroscopy (Fig. S10–S12 and Table S5–S7, ESI†). Thus, a decrease in catalytic reactivity was caused by desorption of Fe complexes in $\text{Fe}^{\text{II}}\text{-}^{\text{Cl}}\text{NHC@Al}^{\text{F}}$ and $\text{Fe}^{\text{II}}\text{-}^{\text{H}}\text{NHC@Al}^{\text{F}}$ from the surface of Al^{F} .

The high recyclability of $\text{Fe}^{\text{II}}\text{-}^{\text{Me}}\text{NHC@Al}^{\text{F}}$ was derived from the δ^+ properties of the methyl groups on the benzimidazole moiety, inducing stronger electrostatic interaction than other complexes; this stronger interaction causes $\text{Fe}^{\text{II}}\text{-}^{\text{Me}}\text{NHC}$ to stay in the hydrophobic micro-environment of Al^{F} . Furthermore, the $\text{Fe}^{\text{II}}\text{-}^{\text{Me}}\text{NHC@Al}^{\text{F}}$ catalyst retained activity up to the 7th run (Fig. S12 and Tables S7, ESI†). Thus, the $\text{Fe}^{\text{II}}\text{-}^{\text{Me}}\text{NHC@Al}^{\text{F}}$ catalyst has recyclability, high 2e^- -oxidation selectivity and a high TON for the CyH oxidation reaction in acetonitrile.

The catalytic oxidation of Cyene was also performed using the three heterogeneous catalysts with H_2O_2 in acetonitrile at 313 K under Ar. Products obtained were analysed by GC-MS analysis. The oxidation products derived from cyclohexene were 2-cyclohexen-1-ol (Cyene-OH), 2-cyclohexen-1-one (Cyene=O) and cyclohexene oxide (Cyene-oxide). The TONs obtained for three catalysts with H_2O_2 are listed in Table 2. In the Cyene oxidation, the $\text{Fe}^{\text{II}}\text{-}^{\text{Cl}}\text{NHC@Al}^{\text{F}}$ catalyst shows the highest total TON of 908 of all the catalysts (Table 2). In contrast, we confirmed that the $\text{Fe}^{\text{II}}\text{-}^{\text{R}}\text{NHC}$ complexes as homogeneous catalysts and the support $\text{H}^+\text{@Al}^{\text{F}}$ showed 100-fold lower catalytic activity and no reactivity in Cyene oxidation with H_2O_2 (Table S8, ESI†). High selectivity for epoxidation as reflected in the ratio (Cyene-oxide/(Cyene-OH + Cyene=O)) was observed for the oxidation of Cyene using $\text{Fe}^{\text{II}}\text{-}^{\text{Cl}}\text{NHC@Al}^{\text{F}}$ as the catalyst. The high selectivity of epoxidation (90%) for $\text{Fe}^{\text{II}}\text{-}^{\text{Cl}}\text{NHC@Al}^{\text{F}}$ suggests that the electron-withdrawing effect of chloro groups can stabilize the $\text{Fe}(\text{II})$ -hydroperoxo ($\text{Fe}^{\text{II}}\text{-OOH}$) species, which should be responsible for the epoxidation.^{12,13}

The redox potentials of $\text{Fe}^{\text{II}}\text{-}^{\text{R}}\text{NHC}$ complexes in CH_3CN were determined by electrochemical measurements. Each complex showed a reversible redox wave due to the $\text{Fe}^{\text{III}}/\text{Fe}^{\text{II}}$ couple at 0.93 V (R = Me), 0.95 V (R = H), and 1.01 V (R = Cl) vs. SCE (Fig. S13 in the ESI†). We also measured the ESR spectra of reaction mixtures of $\text{Fe}^{\text{II}}\text{-}^{\text{Cl}}\text{NHC}$ and H_2O_2 in CH_3CN at 77 K, and there was no ESR signal due to $\text{Fe}(\text{III})$ species (Fig. S14 in the ESI†). In addition, the ESI-MS spectrum of the reaction mixture showed a peak cluster assigned to $[\text{Fe}^{\text{II}}(\text{OOH})(^{\text{Cl}}\text{NHC})]^+$ at $m/z = 570.99$ (Fig. S15 in the ESI†). The results suggest that electron transfer from $\text{Fe}^{\text{II}}\text{-}^{\text{Cl}}\text{NHC}$ to H_2O_2 does not occur to

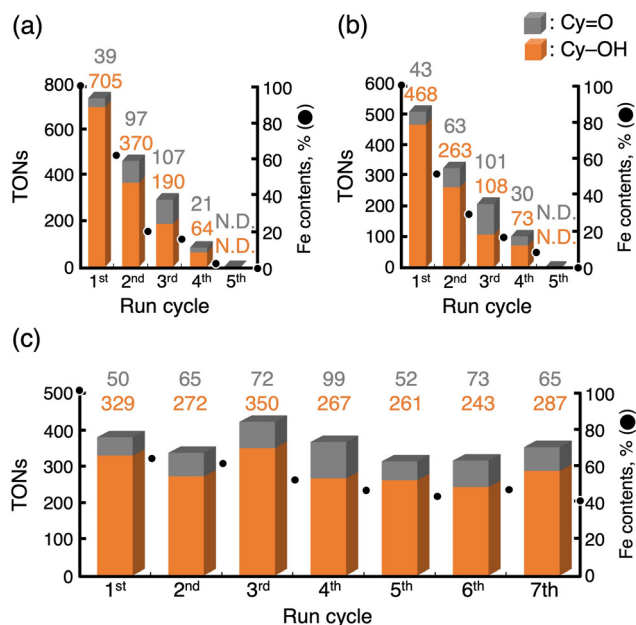


Fig. 2 Total TONs for catalytic CyH oxidation reactions and Fe contents based on the UV-vis-NIR reflectance spectroscopic measurements after each catalytic run using $\text{Fe}^{\text{II}}\text{-}^{\text{Cl}}\text{NHC@Al}^{\text{F}}$ (a), $\text{Fe}^{\text{II}}\text{-}^{\text{H}}\text{NHC@Al}^{\text{F}}$ (b) and $\text{Fe}^{\text{II}}\text{-}^{\text{Me}}\text{NHC@Al}^{\text{F}}$ (c) with H_2O_2 . Conditions: catalyst: 10 nmol, $[\text{H}_2\text{O}_2] = 16 \text{ mM}$, [substrate] = 100 mM, $T = 313 \text{ K}$, and reaction time = 3 h. TONs were determined by GC-MS spectroscopy. No byproducts were observed. N.D.: Not detected. Fe contents, % = $(K/M)_X / (K/M)_0 \times 100$ (X = number of catalytic runs). $(K/M)_X$: The K/M value at 451 nm determined by UV-vis-NIR reflectance measurement of the catalyst after the Xth catalytic run. $(K/M)_0$: The K/M value at 451 nm determined by UV-vis-NIR reflectance measurement of a catalyst before reaction.

Table 2 Catalytic oxidation of cyclohexene by H_2O_2 (30% aq) and $\text{Fe}^{\text{II}}\text{-}^{\text{R}}\text{NHC@Al}^{\text{F}}$ in CH_3CN under Ar

R	Product	TON ^a	Cyene-oxide/ (Cyene-OH + Cyene=O)
Cl	Cyene-oxide	816	8.9
	Cyene-OH	71	
	Cyene=O	21	
H	Cyene-oxide	503	1.6
	Cyene-OH	208	
	Cyene=O	110	
Me	Cyene-oxide	305	2.0
	Cyene-OH	112	
	Cyene=O	42	

^a TONs = [product]/[catalyst]. Conditions: catalysts: 10 nmol, $[\text{H}_2\text{O}_2] = 16 \text{ mM}$, [substrate] = 100 mM, $T = 313 \text{ K}$, and reaction time = 3 h. TONs were determined by GC-MS spectroscopy. No side-reaction products apart from those listed in the table were observed.

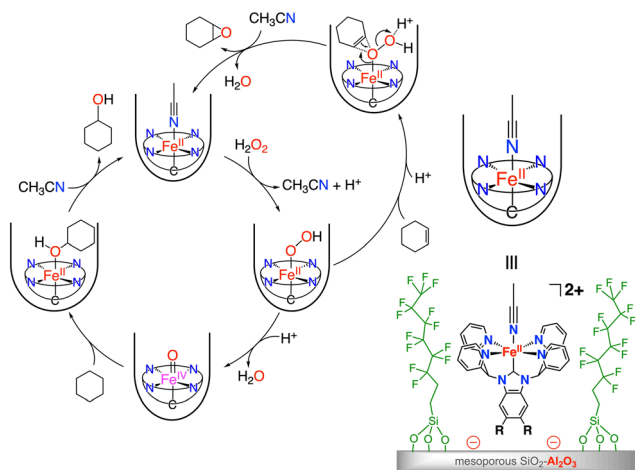


Fig. 3 Proposed mechanisms of the hydroxylation of cyclohexane and epoxidation of cyclohexene by $\text{Fe}^{\text{II}}\text{-R}^{\text{NHC}}\text{@AlF}$ and H_2O_2 .

form Fe^{III} species, but $\text{Fe}^{\text{II}}\text{-Cl}^{\text{NHC}}$ reacts with H_2O_2 to form the $\text{Fe}^{\text{II}}\text{-OOH}$ intermediate. Based on the results of catalytic reactions for CyH and Cyene oxidation, a plausible reaction mechanism is proposed, as shown in Fig. 3. In the first step, an $\text{Fe}^{\text{II}}\text{-acetonitrile}$ complex is converted to the corresponding $\text{Fe}^{\text{II}}\text{-OOH}$ complex through ligand substitution, which is enhanced by the strong *trans* effect of the NHC ligand,¹⁴ by H_2O_2 followed by deprotonation. This $\text{Fe}^{\text{II}}\text{-OOH}$ species can react directly with Cyene to produce Cyene-oxide. As a minor pathway, the $\text{Fe}^{\text{II}}\text{-OOH}$ intermediate undergoes O–O binding heterolysis to afford an $\text{Fe}^{\text{IV}}\text{-oxo}$ ($\text{Fe}^{\text{IV}}\text{=O}$) complex⁶ to perform hydroxylation at the allylic positions *via* hydrogen atom transfer (HAT) to form Cyene–OH, which undergoes over-oxidation to afford Cyene=O. In CyH oxidation, the $\text{Fe}^{\text{II}}\text{-OOH}$ species does not react with the substrate and is converted to the $\text{Fe}^{\text{IV}}\text{=O}$ species, which can hydroxylate CyH through HAT.⁸

In conclusion, we have demonstrated catalytic oxidation reactions of CyH and Cyene by H_2O_2 as an oxidant and $\text{Fe}^{\text{II}}\text{-NHC}$ catalysts immobilized on hydrophobic fluorinated Al-SBA-15 to afford $2e^-$ -oxidized products, Cy–OH from Cy–H and Cyene-oxide from Cyene, respectively, with high TON and selectivity in acetonitrile in comparison with previously reported Fe-based catalysts (Table S9, ESI†).^{3a,12,15} This is the first example of the introduction of a hydrophobic environment on an aluminosilicate surface as a support for catalysts for the preparation of heterogeneous metal-complex catalysts. This work provides a strategy for the introduction of a hydrophobic second-coordination sphere to a transition metal complex as a catalyst without time-consuming and tiresome ligand synthesis. The strategy to construct a hydrophobic micro-environment around a metal complex as a catalyst will be applied to the chemical conversion of various hydrophobic organic compounds in high efficiency and selectivity in polar media.

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Conflicts of interest

There are no conflicts to declare.

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